FIFO Page Replacement Algorithm

Completed

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>

int main() {
    int n, frames_count, i, j;
    int page_faults = 0, page_hits = 0, pointer = 0;

    printf("Enter number of pages in reference string: \n");
    if (scanf("%d", &n) != 1) return 0;

    int reference_string[n];
    printf("Enter the reference string: \n");
    for (i = 0; i < n; i++) {
        scanf("%d", &reference_string[i]);
    }

    printf("Enter number of frames: \n");
    if (scanf("%d", &frames_count) != 1) return 0;

    int frames[frames_count];
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

Input Format

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Output Format

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Sample Input 1

```c
    printf("Enter the reference string: \n");
    for (i = 0; i < n; i++) {
        scanf("%d", &reference_string[i]);
    }

    printf("Enter number of frames: \n");
    if (scanf("%d", &frames_count) != 1) return 0;

    int frames[frames_count];
```

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> Test Case - 1

2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

7

1 3 0 3 5 6 3

3

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Select Language : C (GCC 9.2.0)

```c
    for (i = 0; i < n; i++) {
        scanf("%d", &reference_string[i]);
    }

    printf("Enter number of frames: \n");
    if (scanf("%d", &frames_count) != 1) return 0;

    int frames[frames_count];

    for (i = 0; i < frames_count; i++) {
        frames[i] = -1;
    }

    for (i = 0; i < n; i++) {
        int found = 0;

        for (j = 0; j < frames_count; j++) {
            if (frames[j] == reference_string[i]) {
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

7

1 3 0 3 5 6 3

3

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 6

Total Page Hits: 1

Select Language : C (GCC 9.2.0)

```c
                found = 1;
                page_hits++;
                break;
            }
        }

        if (!found) {
            frames[pointer] = reference_string[i];
            pointer = (pointer + 1) % frames_count;
            page_faults++;
        }
    }

    printf("\nTotal Page Faults: %d", page_faults);
    printf("\nTotal Page Hits: %d\n", page_hits);

    return 0;
}
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

Total Page Faults: 6

Total Page Hits: 1

How it works

```
Ref String | Frames | Status
-----------|--------|-------
1          | 1 - -  | Fault
3          | 1 3 -  | Fault
0          | 1 3 0  | Fault
3          | 1 3 0  | Hit
5          | 5 3 0  | Fault
6          | 5 6 0  | Fault
3          | 5 6 3  | Fault
```

Select Language : C (GCC 9.2.0)

```c
            if (frames[j] == reference_string[i]) {
                found = 1;
                page_hits++;
                break;
            }
        }

        if (!found) {
            frames[pointer] = reference_string[i];
            pointer = (pointer + 1) % frames_count;
            page_faults++;
        }
    }

    printf("\nTotal Page Faults: %d", page_faults);
    printf("\nTotal Page Hits: %d\n", page_hits);

    return 0;
}
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1